AI360

TEAM № 22

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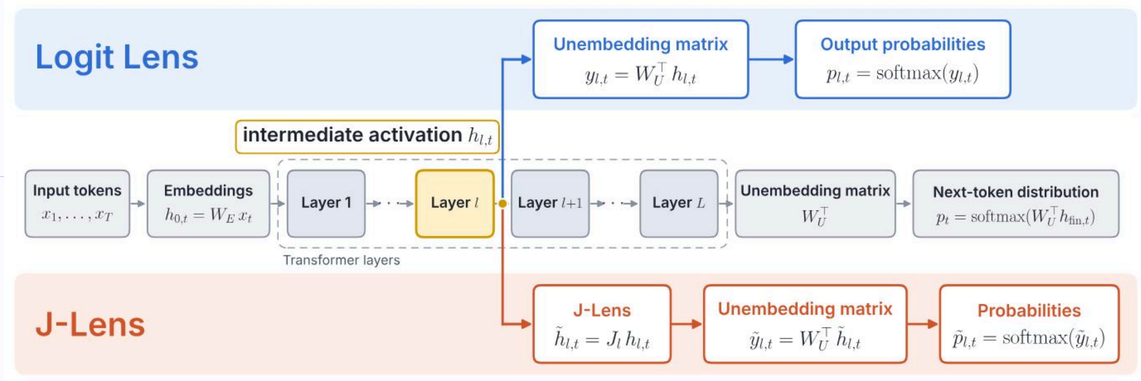
PAPER

Verbalizable Representations Form a Global Workspace in Language Models

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1. Introduction

The logit lens attempts to reveal an LLM's inner workings by applying the unembedding matrix to intermediate-layer activations, but its readouts tend to collapse onto the final prediction and become noisy in early layers. The authors introduce the J-Lens, which measures the “reasoning concepts” encoded in an activation even if they are never explicitly output, and discover the J-Space, an analogue of the neuroscientific global workspace that holds “verbalizable thoughts” shared by many parallel processes.



2. Methods

We aim to measure how an intermediate activation $h_{l,t}$, on average, influences the model’s prediction at future positions $t' \geq t$. For small enough δh,

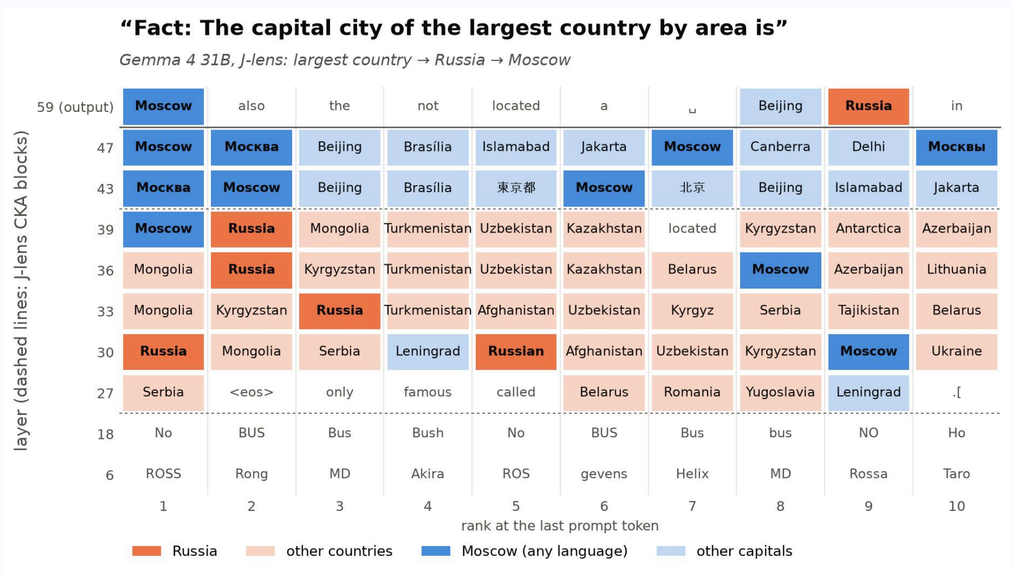
$$\delta h_{\mathrm{fin},t'} \approx \underbrace{\frac{\partial h_{\mathrm{fin},t'}}{\partial h_{l,t}}}_{J_{l,t,t'}} \delta h.$$

Applying the unembedding matrix W_U^T yields $J_{l,t,t'}^T w_{u,i}$, the direction at $h_{l,t}$ that induces the largest increase in the logit of token i at t'. Averaging over prompts and positions to account for context differences, we define

$$J_l \coloneqq \mathbb{E}_{\mathrm{prompt},\, t,\, t' \geq t} \left(J_{l,t,t'} \right), \qquad z_{l,i} \coloneqq J_l^T w_{u,i}$$

as, respectively, the J-Lens and the i-th J-Lens vector for the l-th layer, with the J-Lens readouts defined simply by $W_U^T J_l h_{l,t}$. Since such a system of vectors is overcomplete, we restrict ourselves to nonnegative sparse linear combinations, that is for a given k we define the J-Space of the l-th layer as

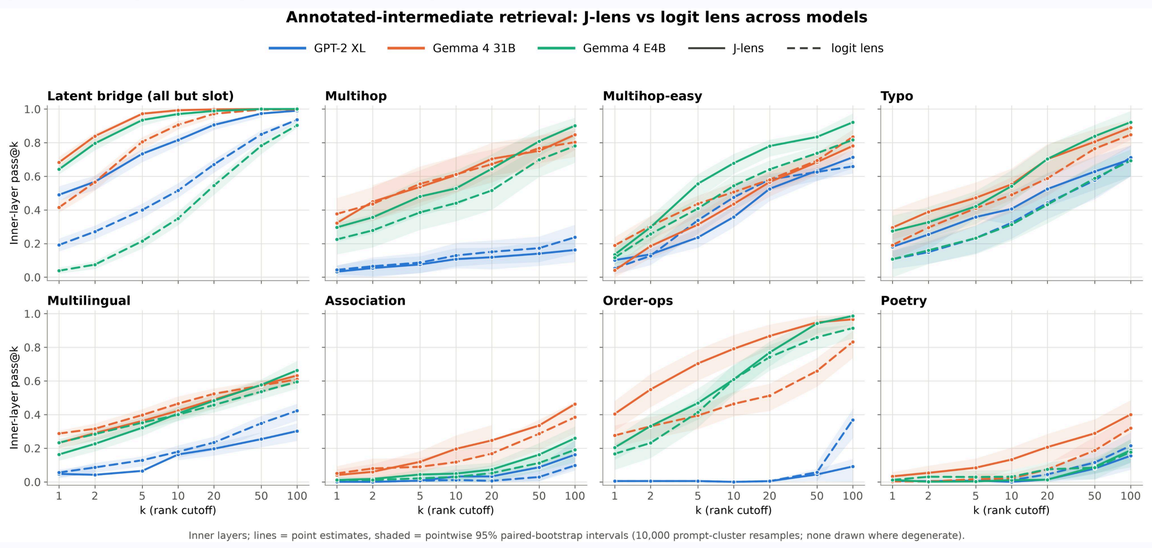
$$\bigcup_{\substack{I \subseteq \{1, \ldots\} \\ |I| \leq k}} \left\{ \sum a_i z_{l,i} \mid a_i \geq 0, i \in I \right\}.$$



3. Experiments

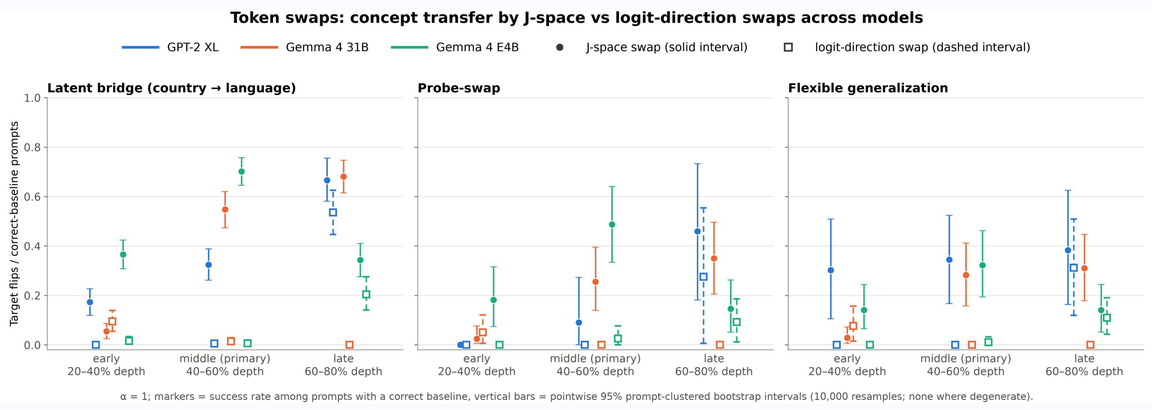
3.1. Intermediate Retrieval

We test the J-Lens' capacity for intermediate concept retrieval by measuring the average percentage of concepts ranked ≤k at any position over a dataset of pairs (prompt, concepts needed to answer).



3.2. The J-Space mediates internal reasoning

To test whether the former result reflects a causal relationship, we measure how frequently swapping two J-Lens vectors changes the model’s answer to a given question, and compare this with the effect of swapping the corresponding logit-lens directions.



3.3. How does the J-Space evolve across layers?

We use centered kernel alignment to compare the J-Spaces across layers. The resulting matrix has explicit block structure, and in newer models the transition between blocks corresponds to a change in intermediate surfacing.

